

Engineering portfolio

Thomas Lepère

M. SC. In Mechanical Engineering - Fluid Mechanics specialization

CFD | Aerodynamics | Wind tunnel testing | Data analysis

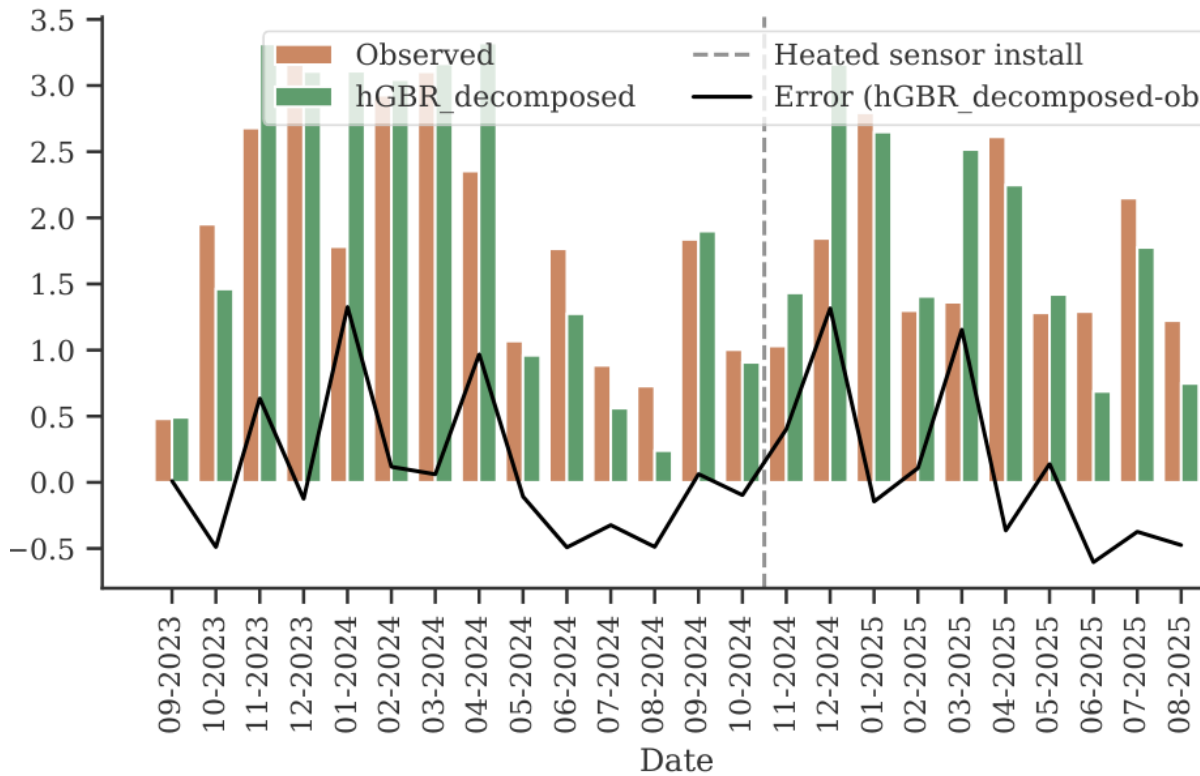
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Estimation of the energy losses from wind and icing predictions on a wind turbine

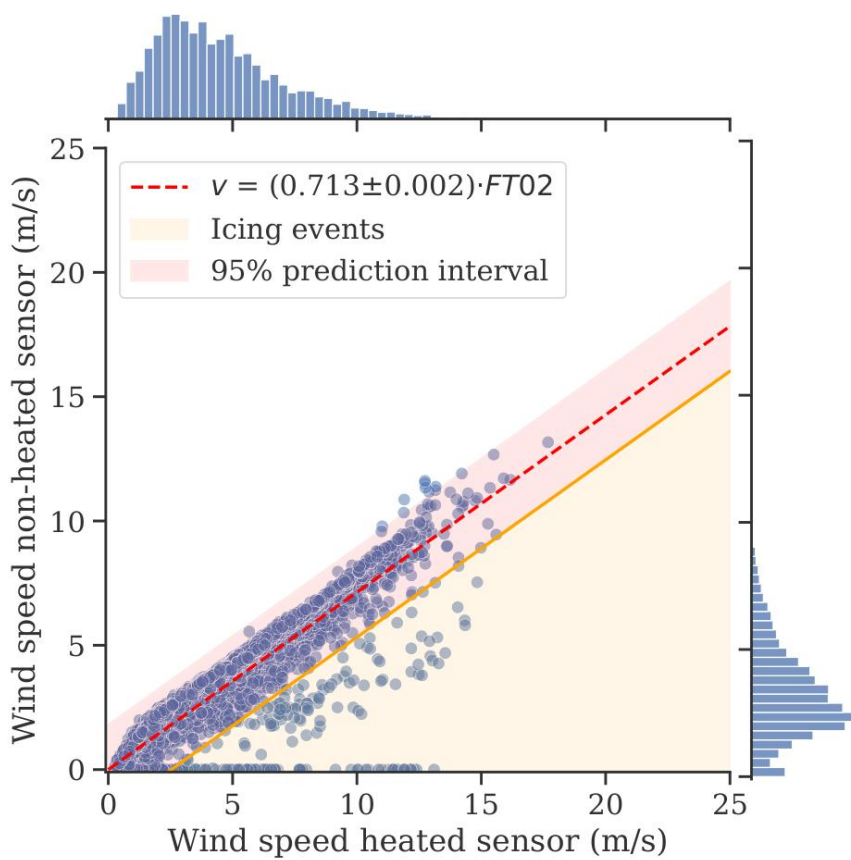
Semester project – CRYOS laboratory

Python

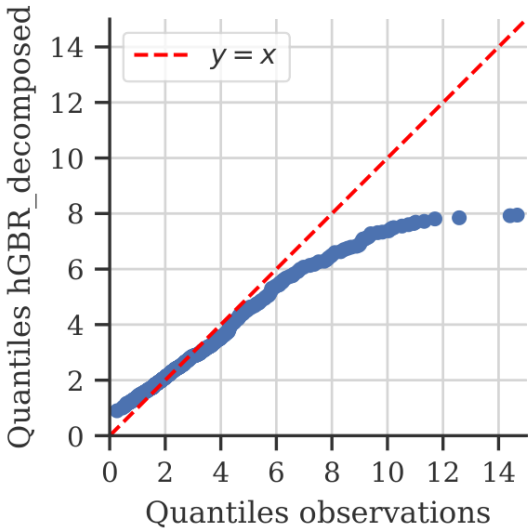
Wind turbine – Data analysis



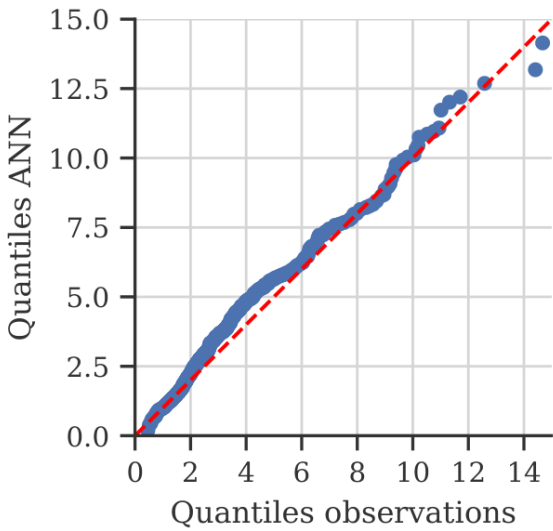
Energy prediction vs observed since the turbine exists



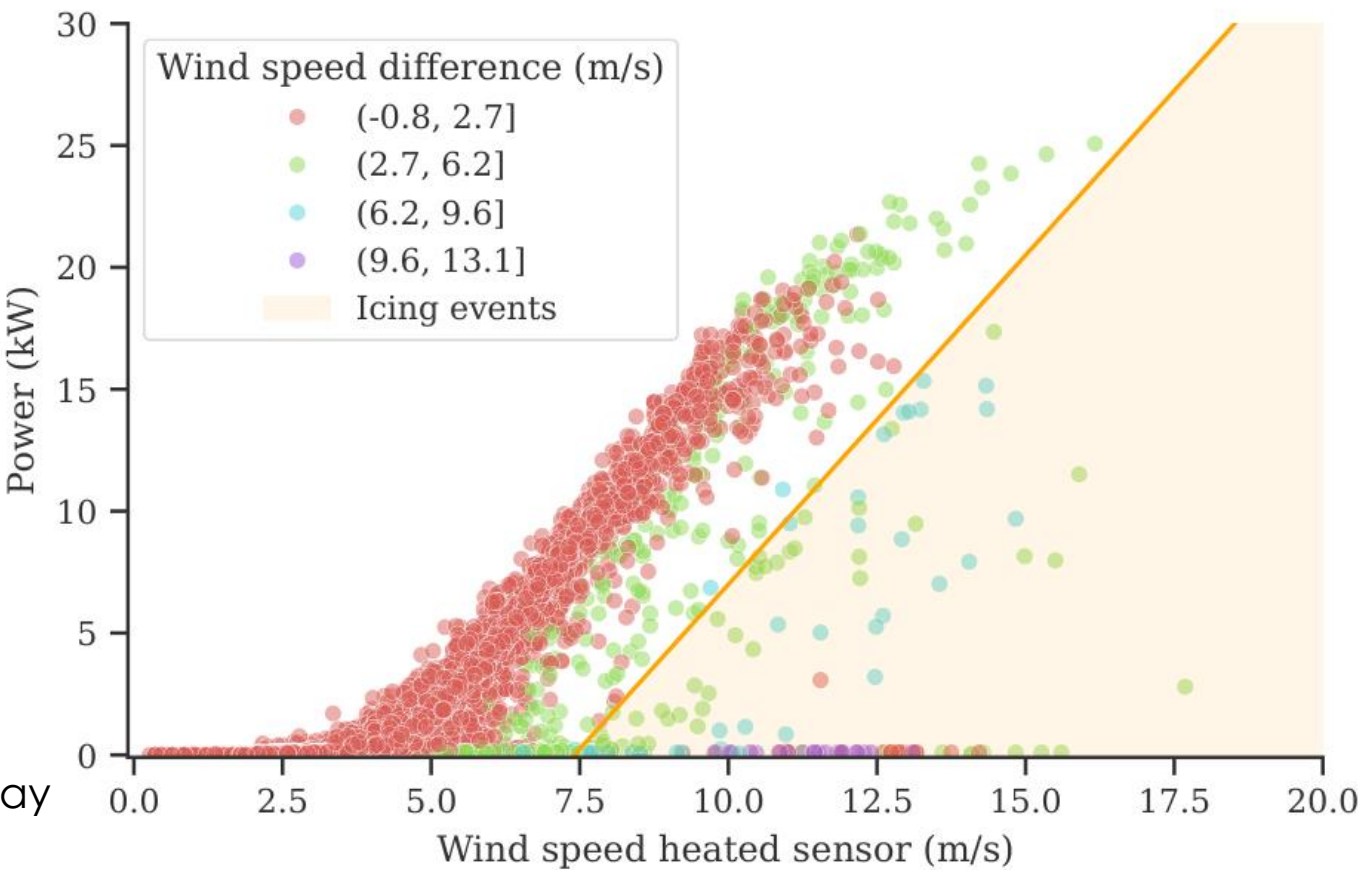
Icing types 1 & 2



(a) q-q plot between observations and hGBR_decomposed



(b) q-q plot between observations and ANN



Icing type 3

Mission quantify production losses due to icing and assess long term wind potential

Method labeled icing from heated/non-heated sensors and trained ML models for predictions

Results 272 icing events, 5-18% production losses and underestimated annual energy production

98% icing prediction accuracy

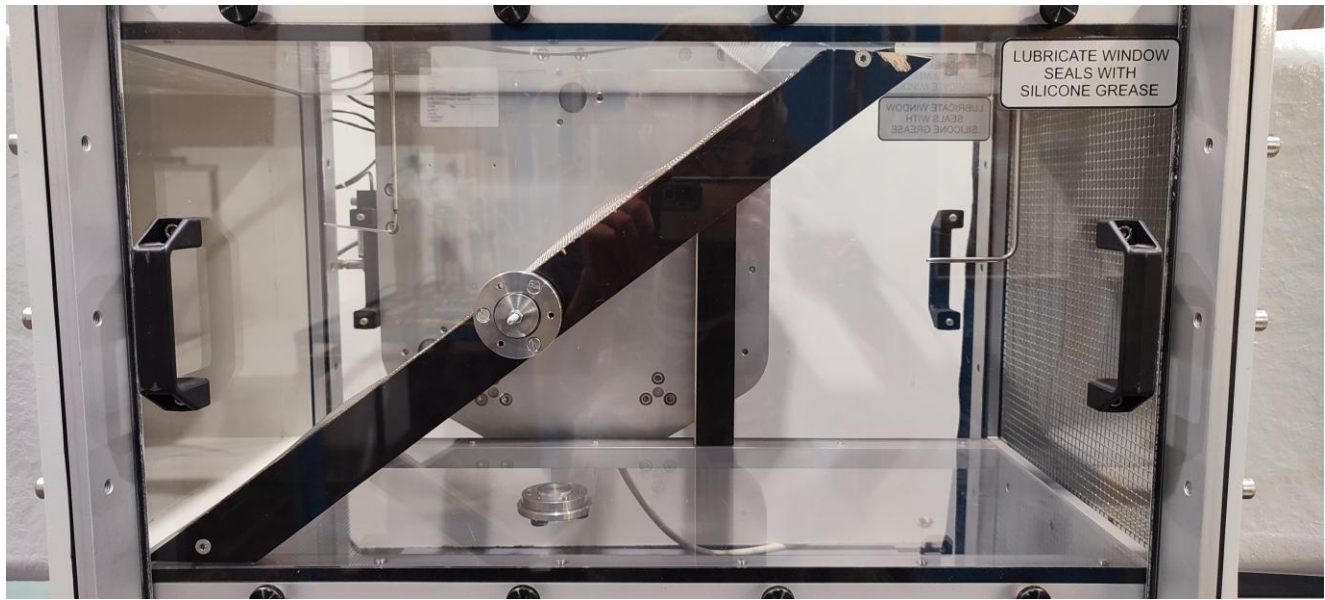
2.2-7.6 MWh energy lost

CFD correlation of a foiling boat

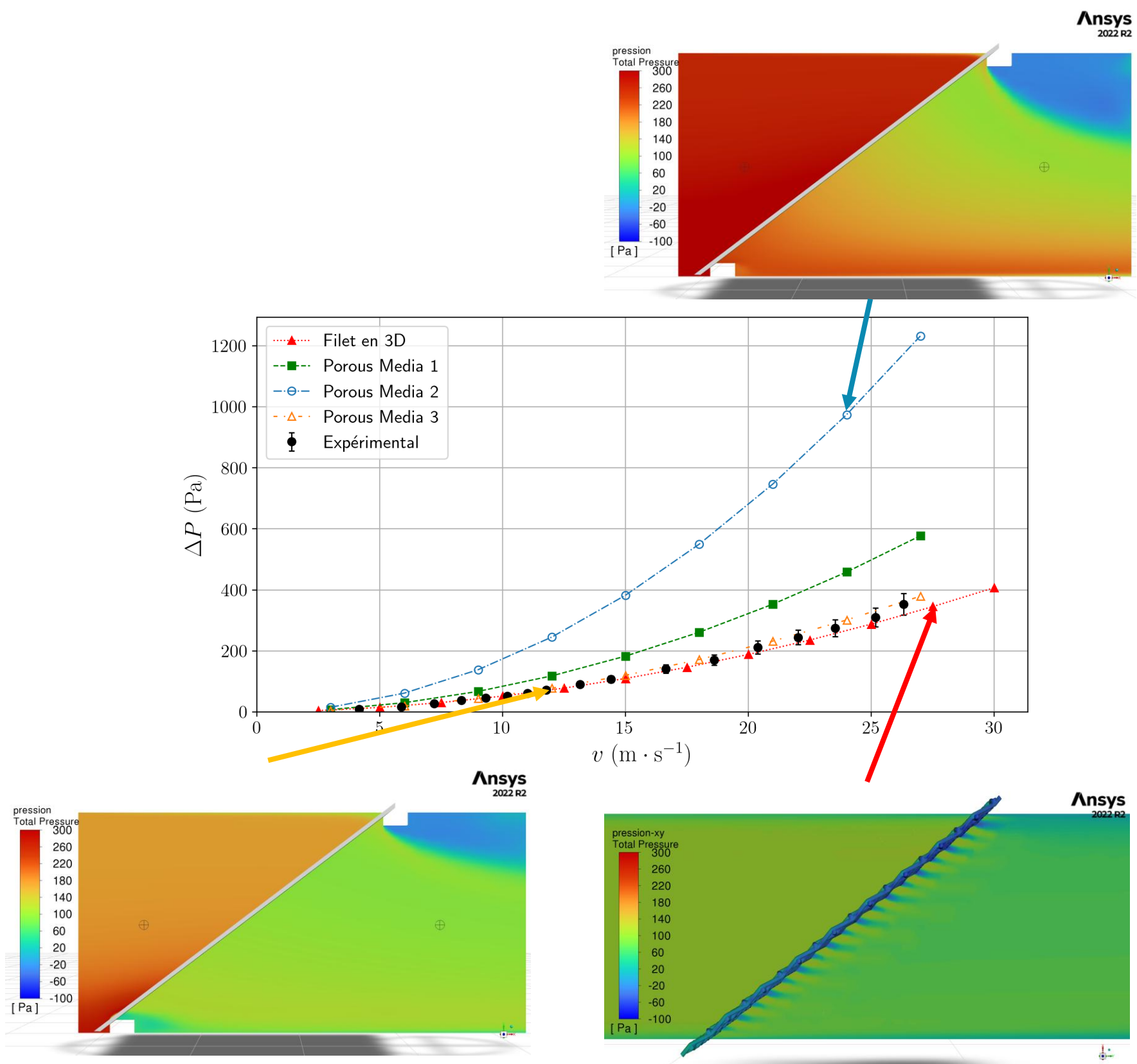
Bachelor thesis - HEPIA

ANSYS Fluent, Python

Aerodynamics – Wind tunnel testing – CFD - Correlation



- Aim** understand the mismatch between previous wind tunnel and CFD drag predictions
- Work** recreated reduced-scale wind tunnel conditions, compared porous-model strategies and pressure losses
- Result** reproduced the pressure loss of a non-perpendicular incident flow but not the real behavior -> not the good way to model the net

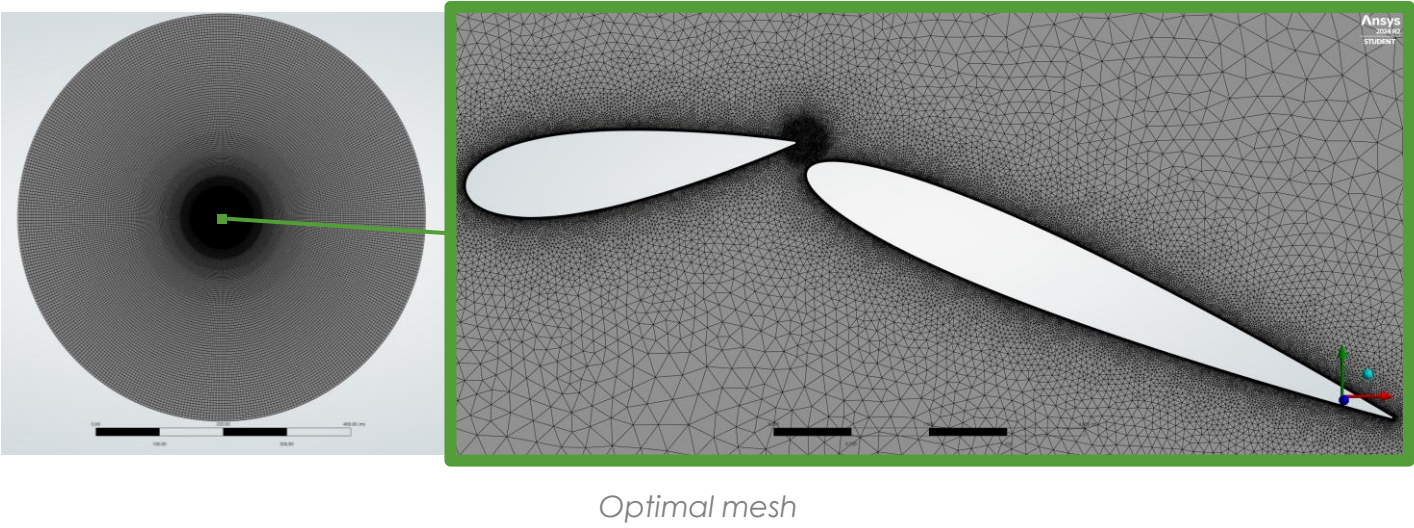
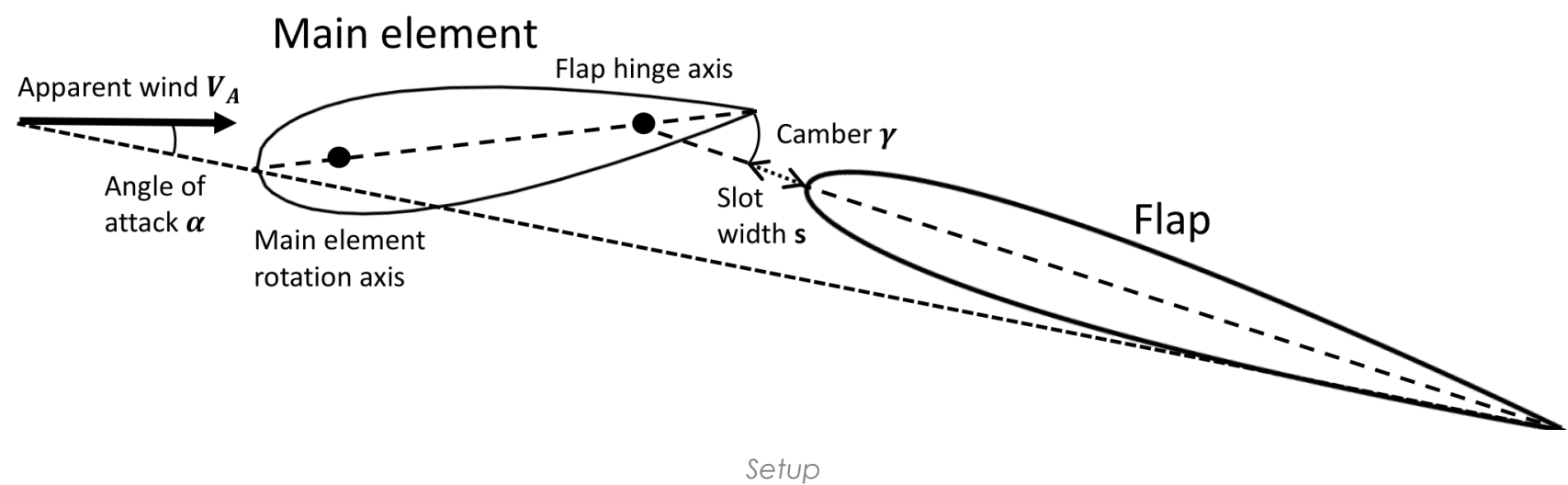


On the effect of slot width on two-elements wingsails

Course project – Numerical Flow Simulation

ANSYS Fluent, Python

Aerodynamics – CFD



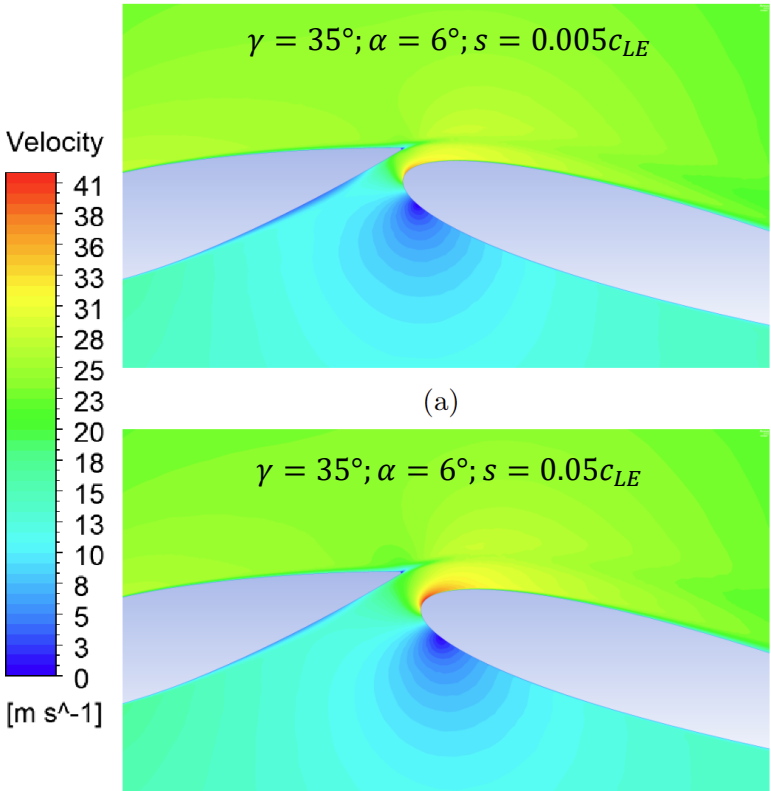
0.5-5% slot width $Re \approx 10^7$ turbulent regime

k- ω SST turbulence model

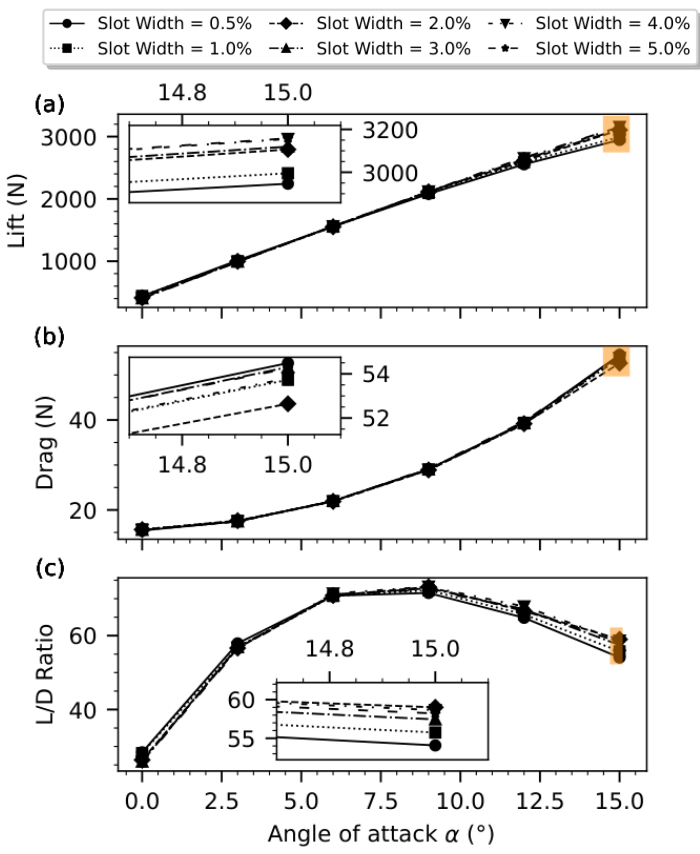
Question when does the slot help or hurt wingsail performance ?

Method varied angle of attack, camber and slot width after mesh/domain convergence

Result slot performance depends strongly on configuration; larger gaps benefit high-angle cases



Contour plot of the velocity magnitude



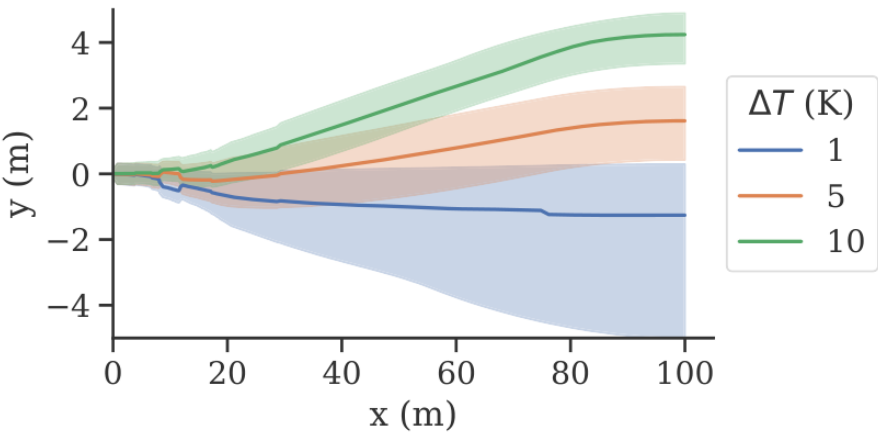
Simulation results showing lift (a), drag (b), and lift-to-drag ratio (c) for various slot widths at a camber angle of 15°

Simulation of Particle-Laden Thermally Buoyant Horizontal Jets

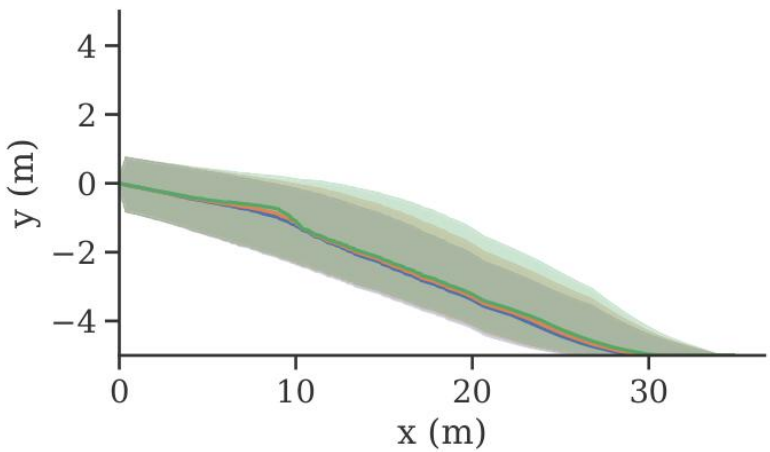
Course project – Environmental Transport Phenomena

ANSYS Fluent

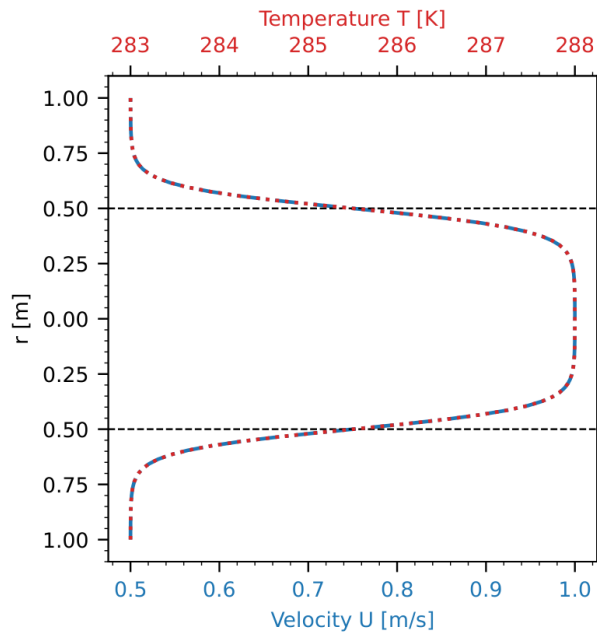
Environmental engineering – CFD



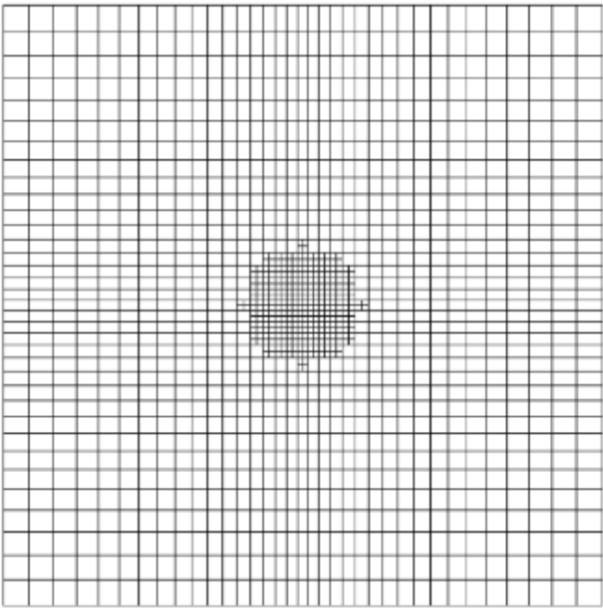
Trajectory of the thermal plume



Trajectory of the sediment plume centerlines



Inlet temperature and velocity profiles



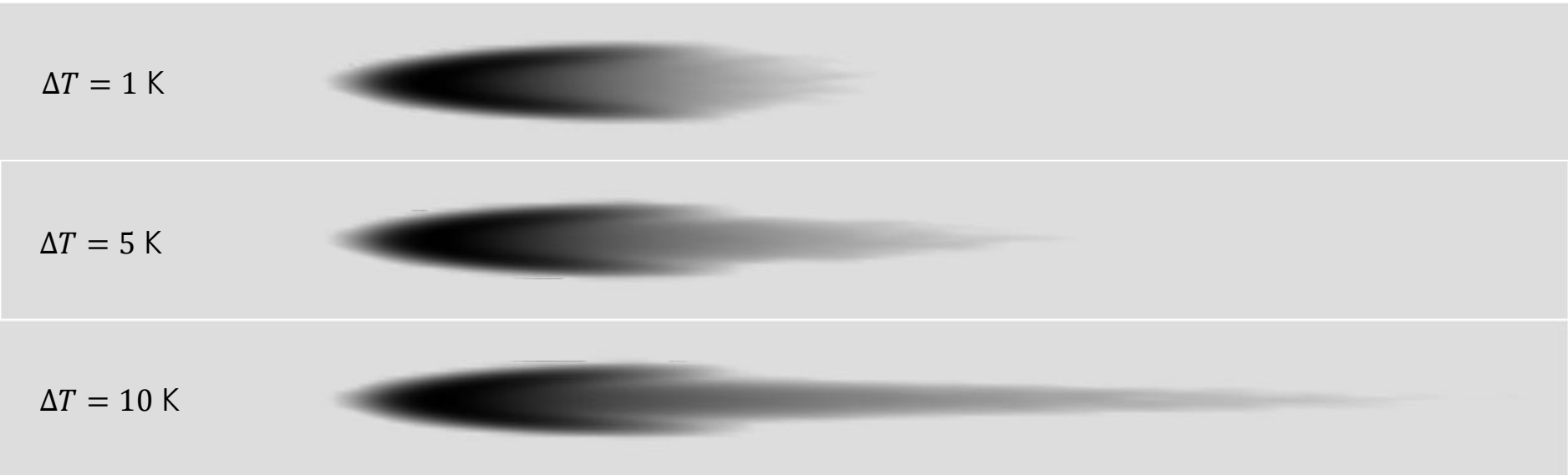
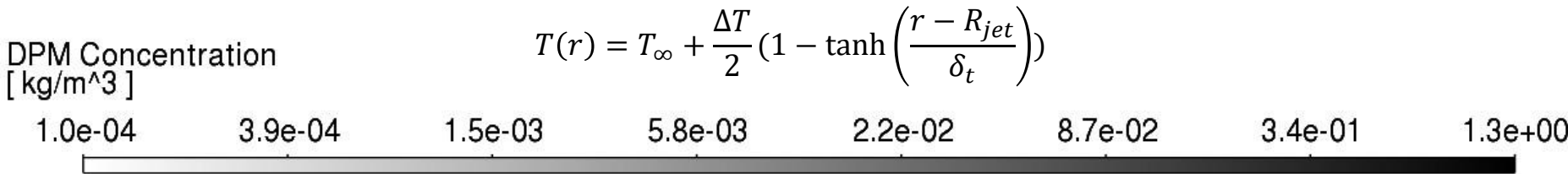
Optimal mesh at the inlet

5x10⁴ particles DPM

Aim simulate how buoyancy changes jet development and particle deposition

Method 3D steady RANS carrier phase with Lagrangian DPM particles and mesh convergence

Result temperature strongly bends the plume; deposition changes mainly for small particle fraction



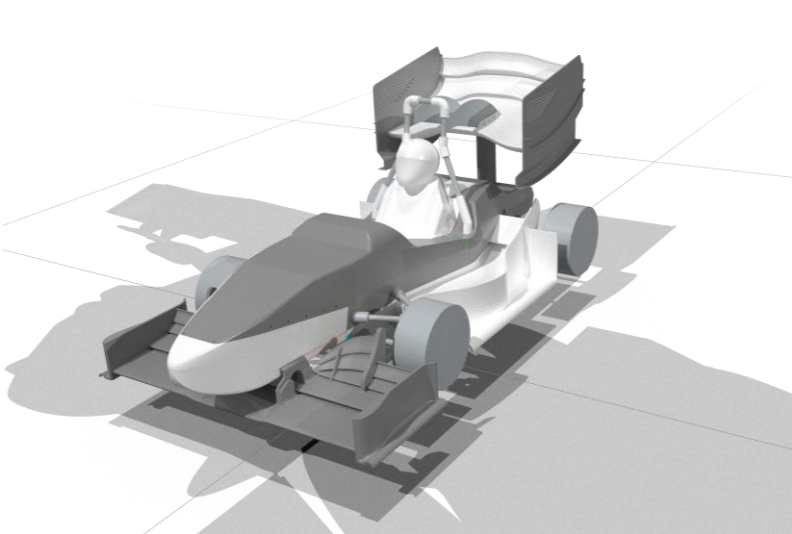
Sediment deposition maps for different jet temperature

CFD correlation of a Formula Student

Semester project – EPFL Racing Team

3DExperience, 3D printing, Python

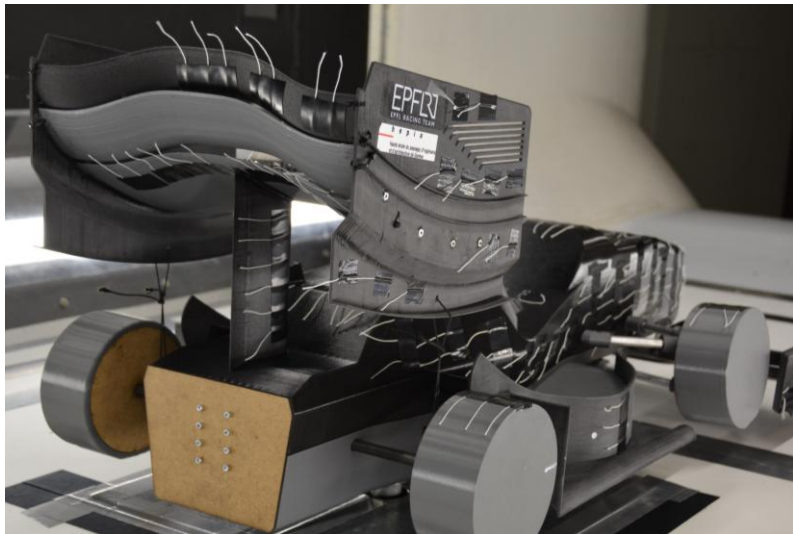
Aerodynamics – Wind tunnel testing – CAD - Correlation



CAD of the reduced model



Photography of real car's rear wing



Reduced model flow at 90 km/h



Reduced model in the wind tunnel

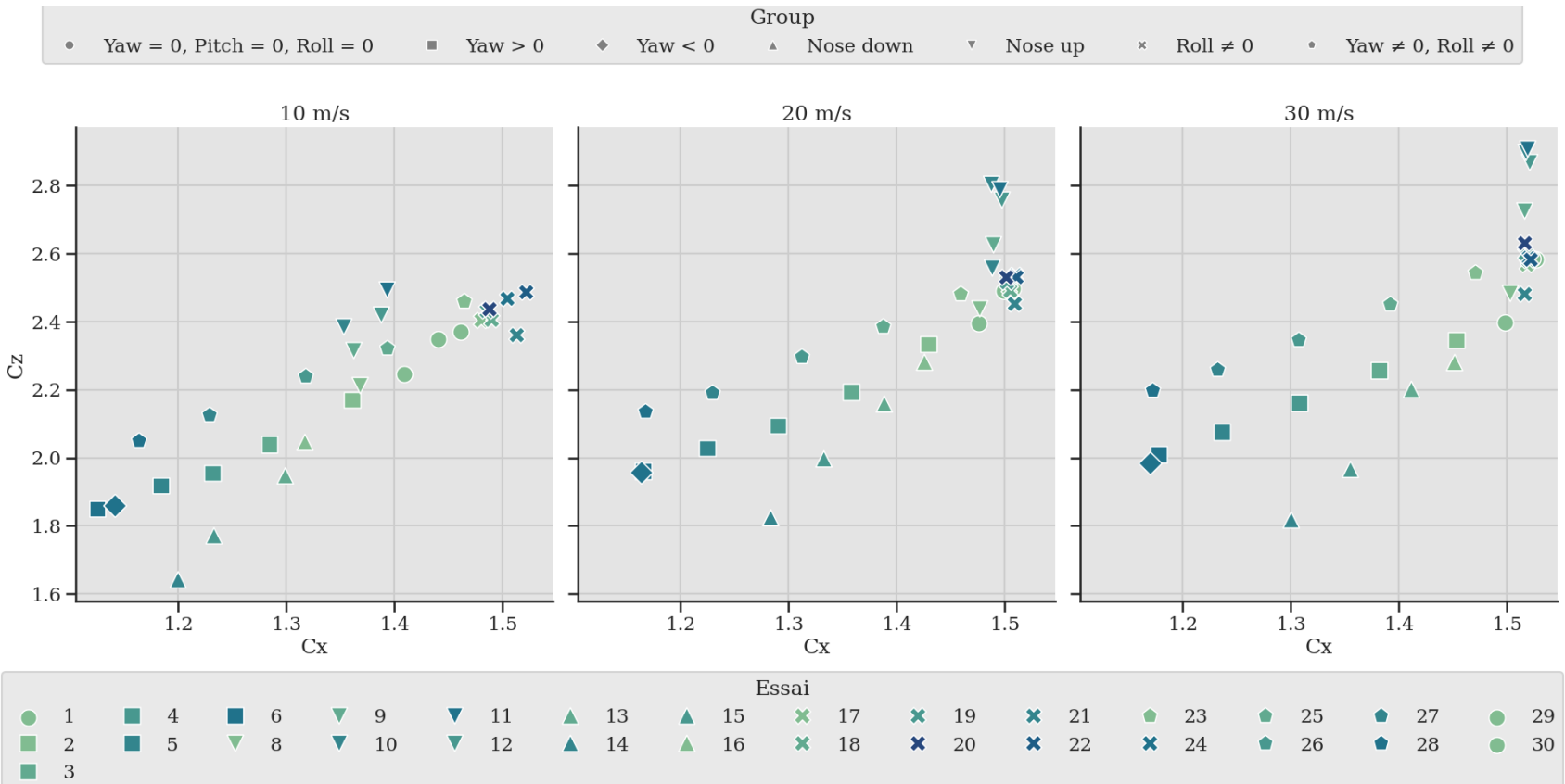
149 tests done

50 m/s maximum wind speed

Objective validate the CFD methodology

Method compare wind tunnel measurements/observations with CFD using a 1/3, 3D printed reduced model of the car

Results less than 10% error for drag coefficients, lift is more complicated to correlate



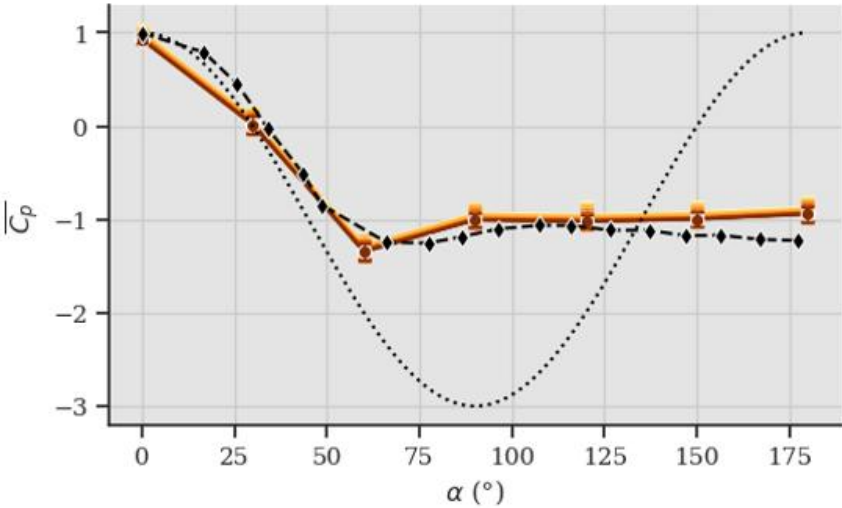
Results of the wind tunnel tests for 3 different wind speeds

Influence of pneumatic tube length on pressure measurements

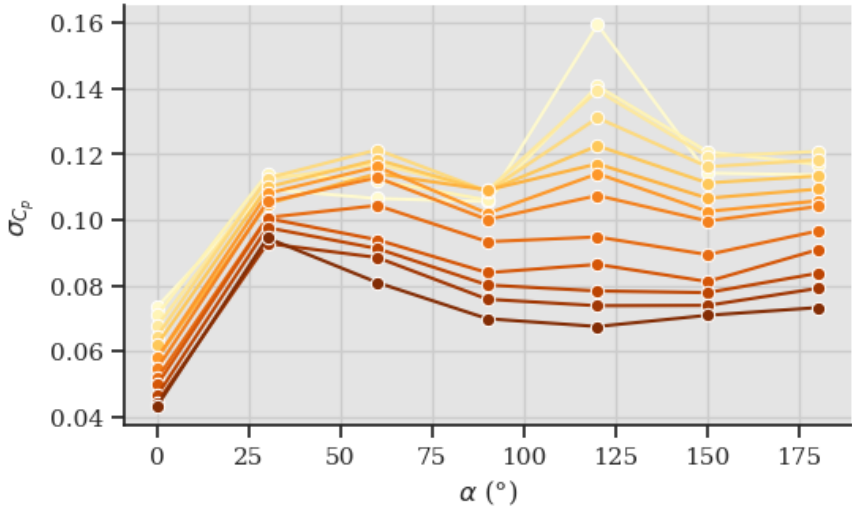
Semester project - UNFoLD laboratory

Python

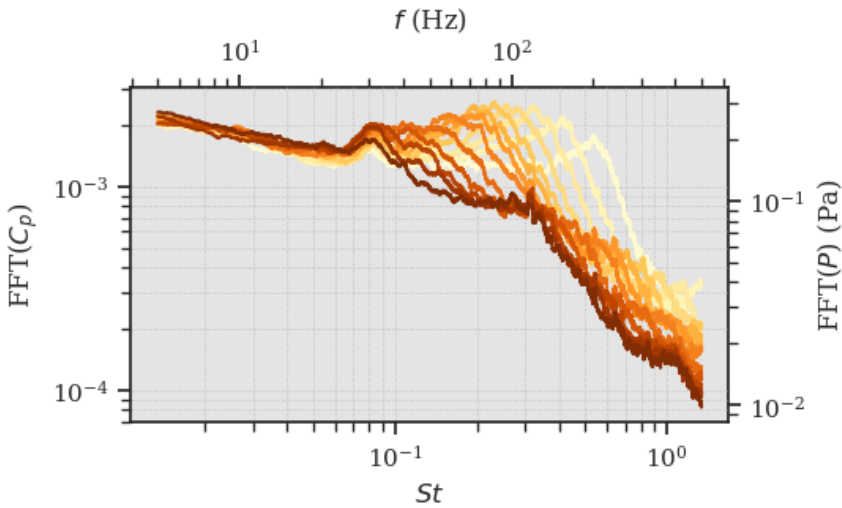
Aerodynamics – Wind tunnel testing



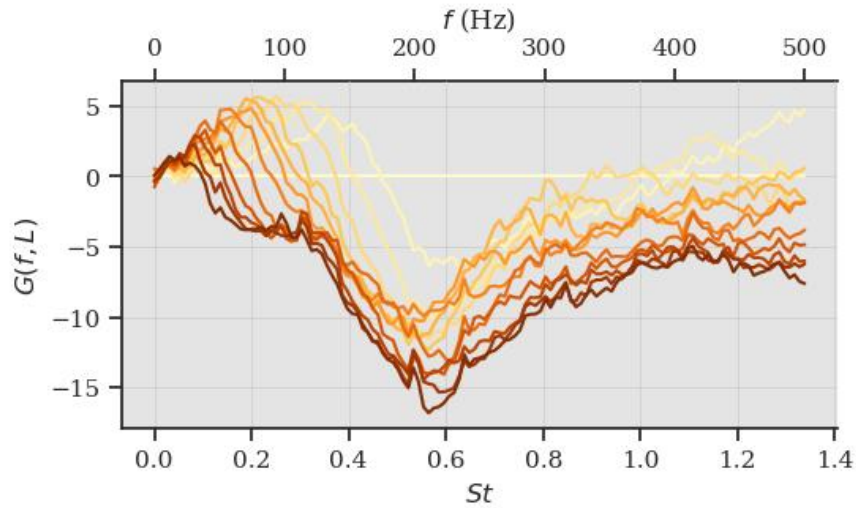
Average pressure coefficient



Standard deviation of pressure coefficient



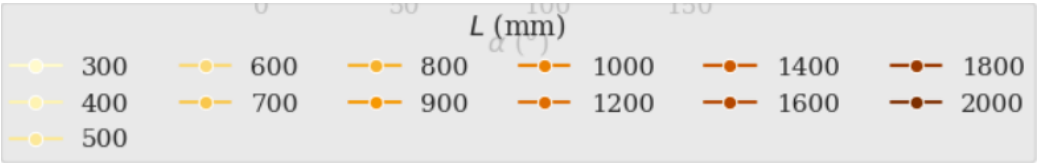
Fourier Transform of the pressure signal for AoA=150°



Spectral gain for AoA=150°

13 tube lengths tested

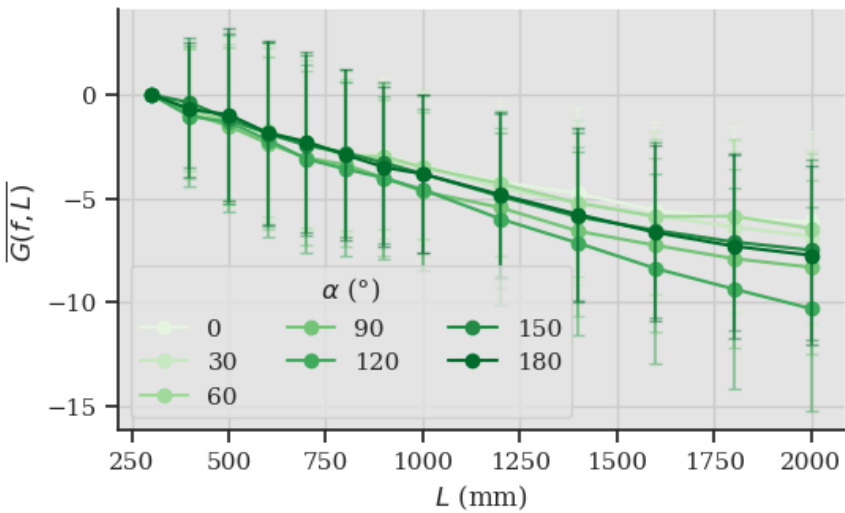
36% fluctuation attenuation



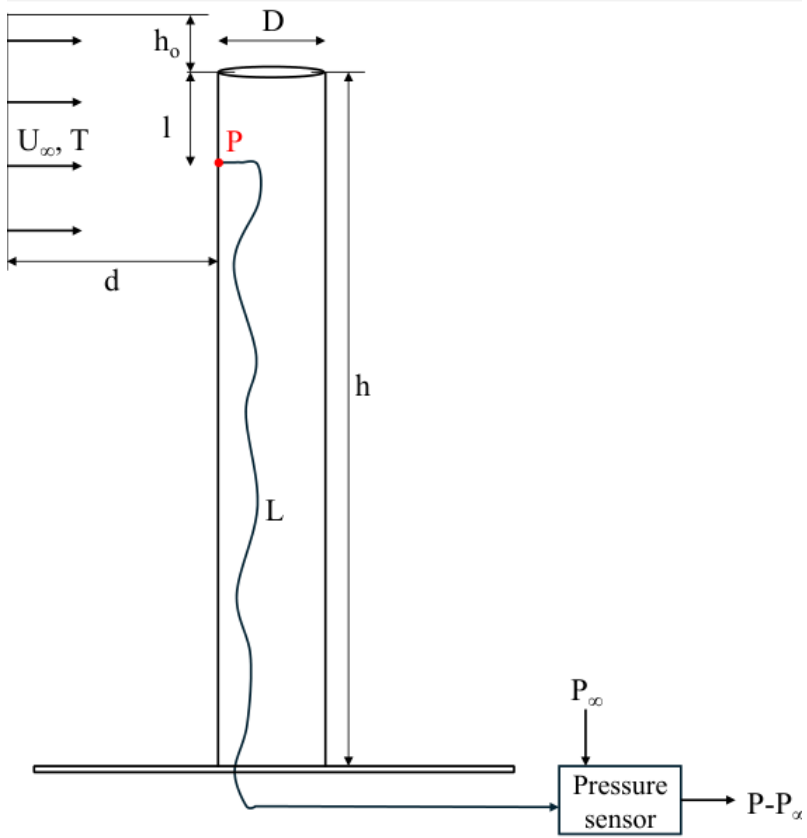
Question what is the impact of pneumatic tube length on pressure measurement ?

Finding mean pressure remains robust; unsteady amplitudes and spectra are strongly affected

Skill designed an experiments, processed statistics and performed spectral analysis



Average spectral gain for each AoA and tube length



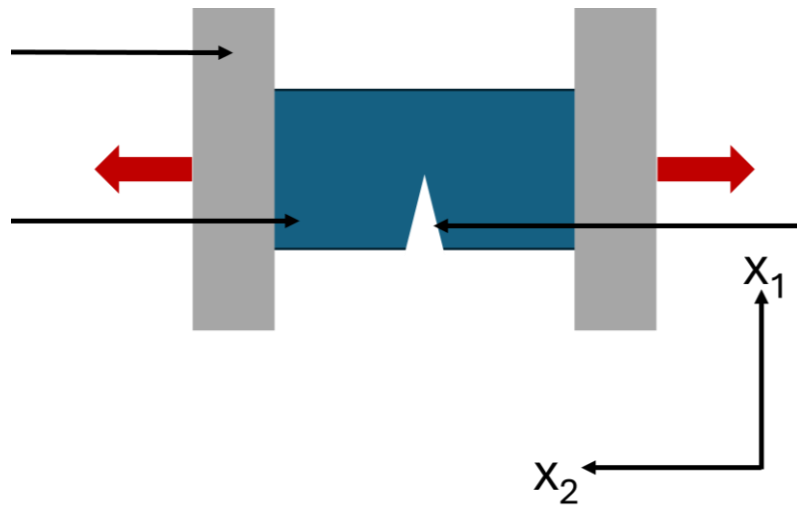
Experimental setup

Investigation on the deformation field around a crack tip using Digital Image Correlation (DIC)

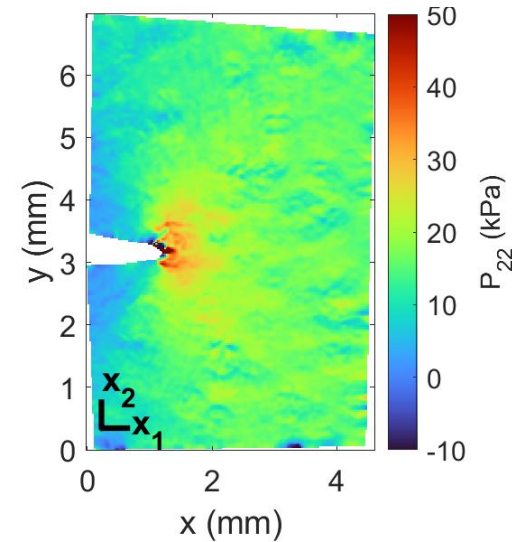
Course Project – Experimental Methods in Mechanical Engineering

Matlab

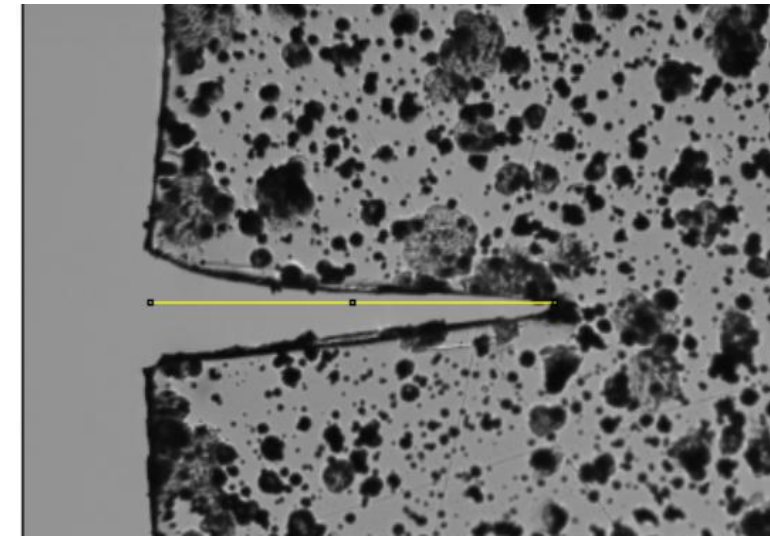
Digital Image Correlation - Structural mechanics



Experimental setup



P_{22} of the Piola-Kirchoff stress tensor



Crack size

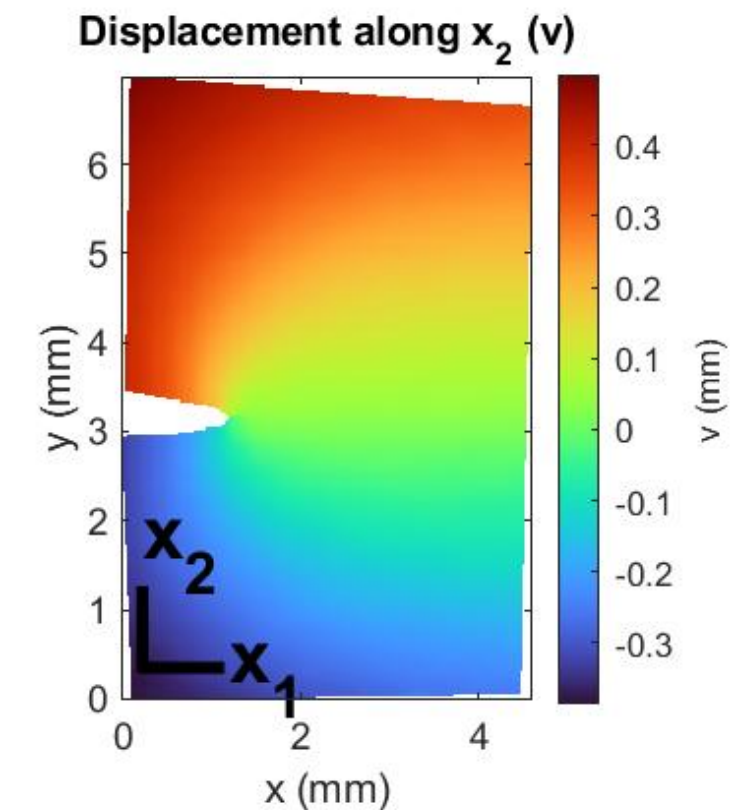
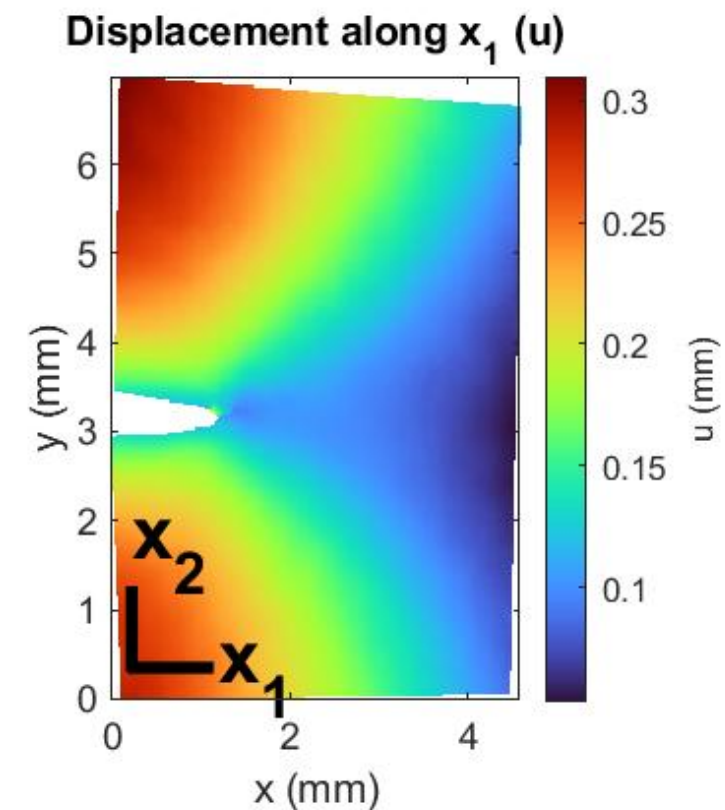
10 frames per second

20 $\mu\text{m/s}$ traction speed

Objective Assess the validity of linear elastic fracture mechanics for characterising crack propagation in highly deformable polyacrylamide hydrogels

Method analysed tensile tests using Digital Image Correlation to determine displacement and strain fields, then calculated the J -integral and compared K_{linear} with K_{plastic}

Results linear model accurately represented two samples, while the third showed a significant discrepancy, highlighting the influence of material nonlinearity and experimental conditions

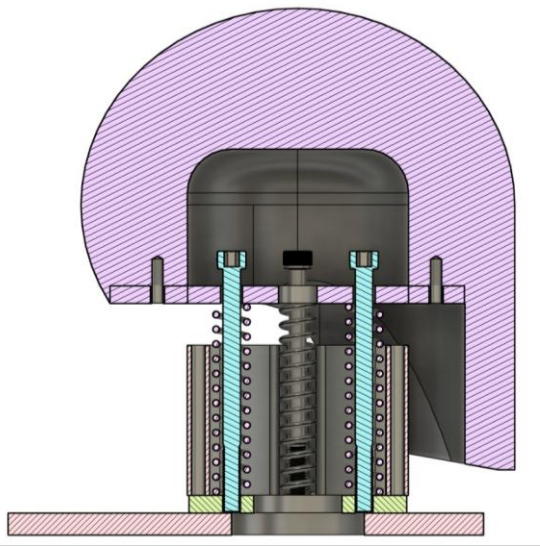


Improving ski helmet testing using a biofidelic neck model

Course project – Biomechanics of the Musculoskeletal system

Autodesk fusion, Python

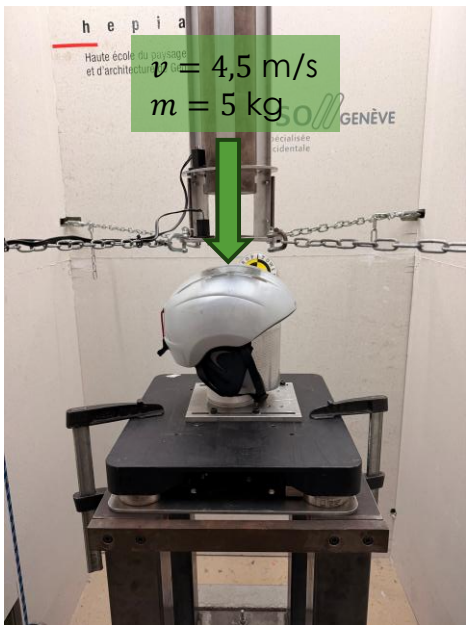
Biomechanics – Crash - CAD



Design of the neck



Photography of the setup



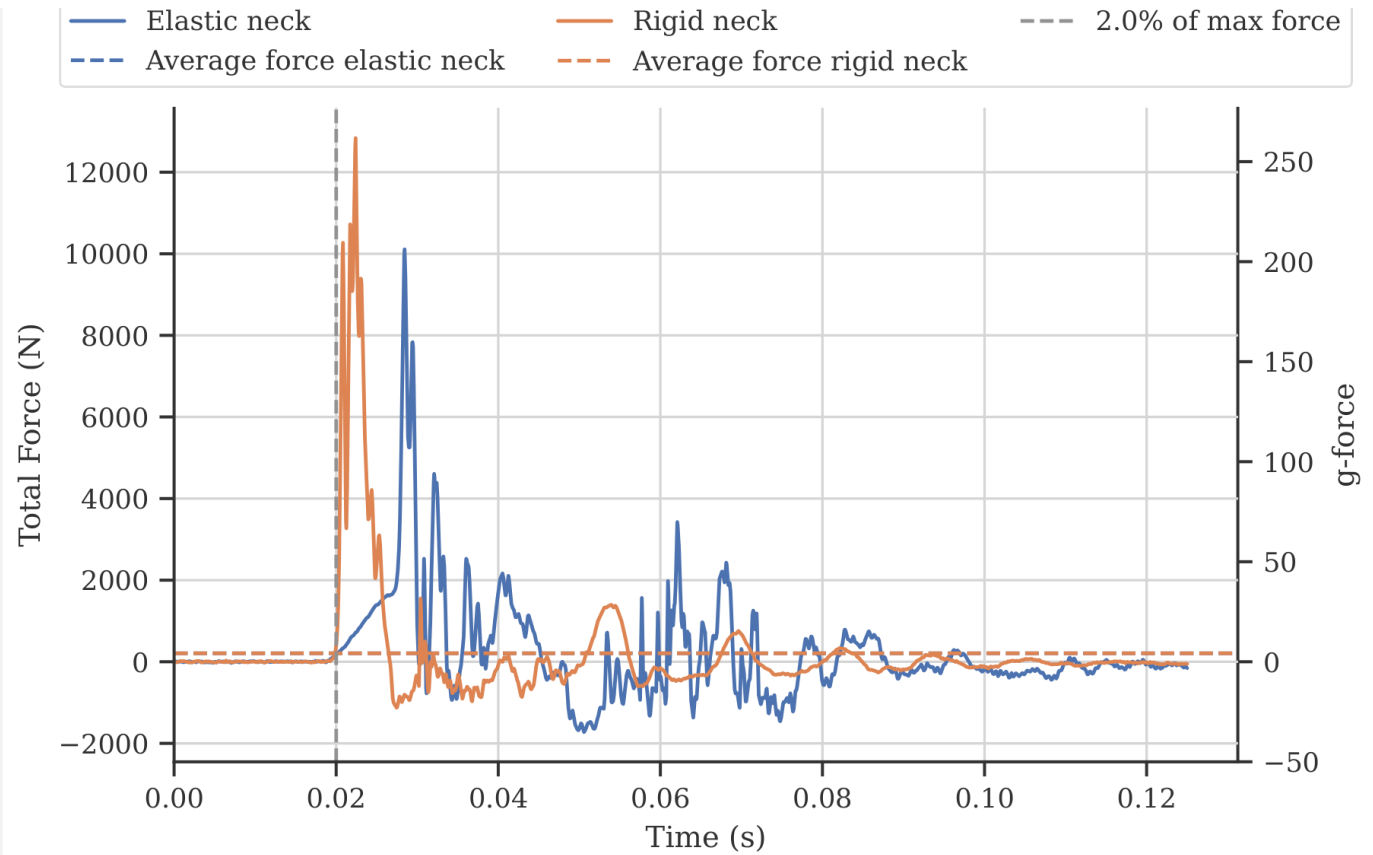
Experimental conditions

19.2 Hz sampling **12000 N** maximum compression

Problem standard helmet tests often use and isolated rigid head and miss neck compliance

Work designed a spring-based, manufacturable neck prototype with-stop and guide éléments

Test drop-tower impacts compared rigid and compliant configurations using force measurements



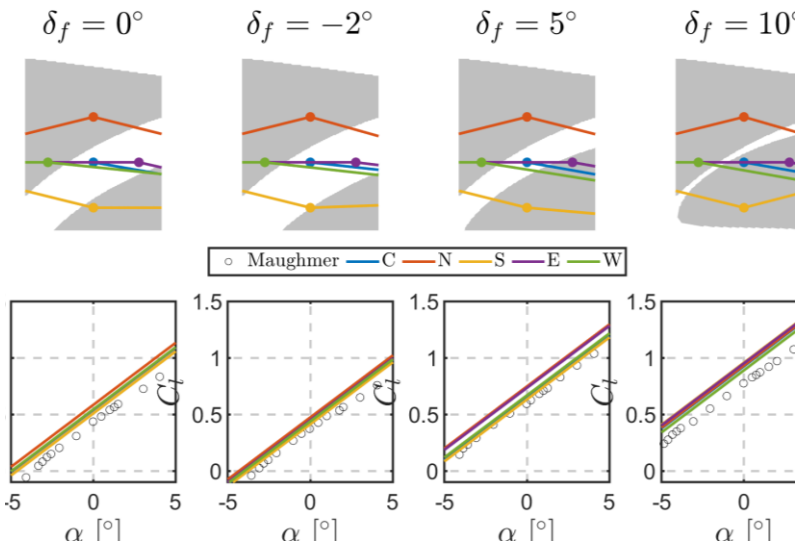
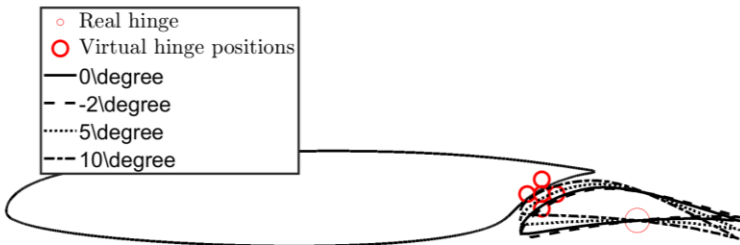
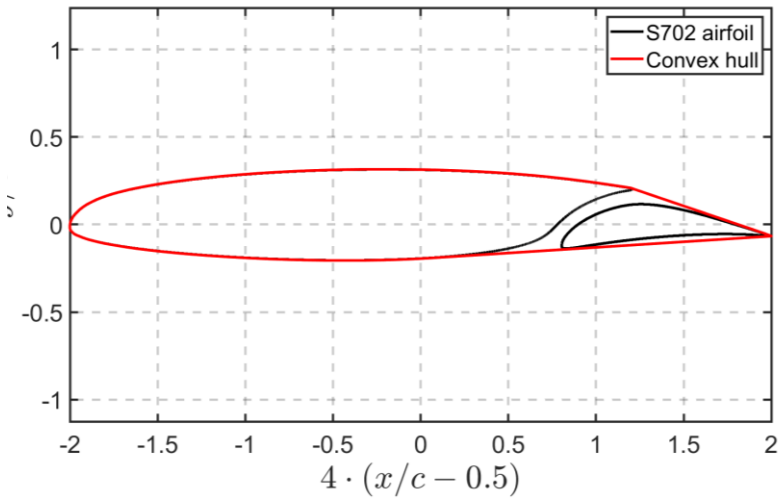
Forces measured by the table with and without the neck

Computation of lift and drag of a 2D airfoil

Course project - Aerodynamics

MATLAB

Aerodynamic – Thin Airfoil Theory – Potential flow theory

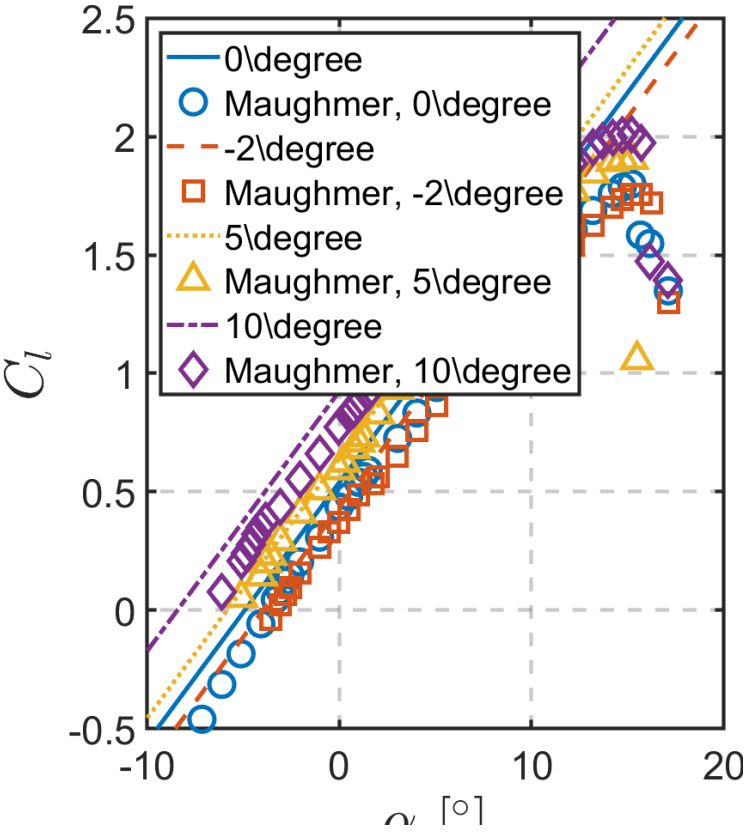
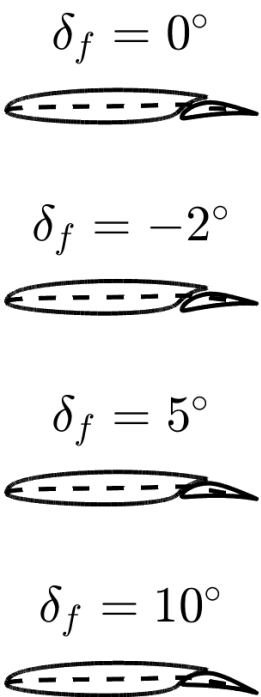


5 different hinge positions

4 flap angles

Problem reproduce the lift and pressure distributions of a 2D airfoil from a paper

Method computation of lift using Thin Airfoil Theory with an arbitrary-defined hinge position for the flap

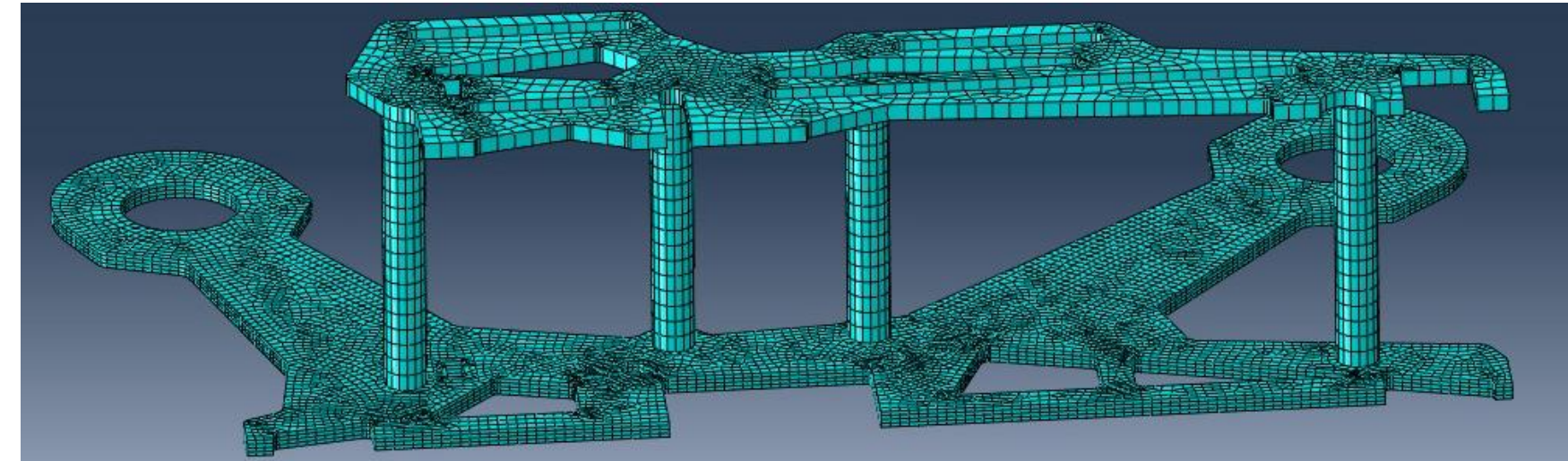
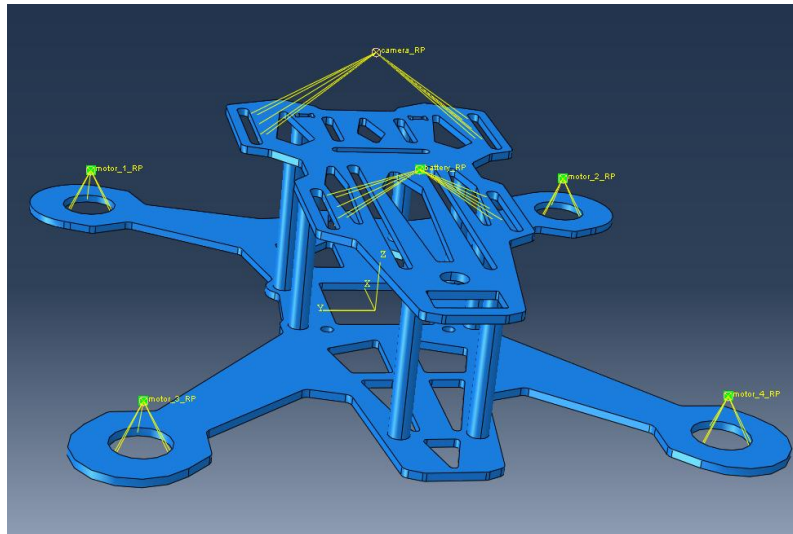


Structural simulation of a quadcopter drone

Course project – Méthodes des Elements Finis

Abaqus, Python, Excel

Structural simulation



4 cas de charge

Objective validate the structural resistance of an FPV drone frame and identify opportunities to reduce its weight and improve flight endurance

Method performed modal and static finite element analyses with mesh-convergence studies for hovering, camera integration and asymmetric turning conditions

Results the frame withstands the modelled normal-flight conditions with a substantial structural margin, supporting further weight reduction through topology optimisation or alternative materials

